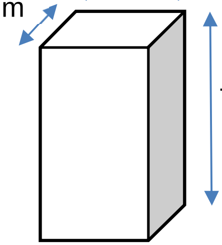
6	The diagram below shows a wooden block in its upright position, with length 1.0 m, width 3.5 m and height 7.0 m. The density of wood is 850 kg m^{-3}.  What is the submerged depth of the wooden block when it floats upright in water? Density of water is 1000 kg m^{-3}.

	A	0.85 m	**B**	5.6 m	**C**	6.0 m	**D**	7.0 m

Answer: C

Weight of water displaced = weight of wooden block

$\rho_{water}\ g\ V_{water\ displ.} = \rho_{wood}\ g\ V_{wood}$

$1000 \times g \times h_{water\ displ.} \times A = 850 \times g \times 7.0 \times A$

$h_{water\ displ.} = 5.95\text{ m} = 6.0\text{ m (2 s.f.)}.$